Mathematical Rigor and Cybernetic Validation of Stylometric Lineage Tracking within OSSL Ecosystems

Metric Tensor Optimization and Invariant Sensitivity Analysis

Fisher Information Matrix and Stylometric Manifold Geometry

Tracking model lineage across open-source software and licensing (OSSL) ecosystems requires mapping high-dimensional textual outputs to an intrinsic representation space. Let $\mathcal{X}$ denote the formal space of generated text sequences. Define a continuous feature extractor function $\mathbf{\Psi}: \mathcal{X} \to \mathbb{R}^D$ that maps a text sample $x \in \mathcal{X}$ to a raw stylometric state vector $\mathbf{s}(x) = \mathbf{\Psi}(x) \in \mathbb{R}^D$. To ensure standard scale invariants across heterogeneous feature domains, the normalized z-score feature vector $\mathbf{z}(x) \in \mathbb{R}^D$ is defined as:

$$\mathbf{z}(x) = \mathbf{\Sigma}_0^{-1/2} (\mathbf{s}(x) - \boldsymbol{\mu}_0)$$

where $\boldsymbol{\mu}_0 = \mathbb{E}_{x \sim \mathcal{D}_0}[\mathbf{s}(x)] \in \mathbb{R}^D$ and $\mathbf{\Sigma}_0 = \mathbb{E}_{x \sim \mathcal{D}_0}[(\mathbf{s}(x) - \boldsymbol{\mu}_0)(\mathbf{s}(x) - \boldsymbol{\mu}_0)^T] \in \mathbb{R}^{D \times D}$ denote the empirical mean vector and covariance matrix computed over a reference text corpus $\mathcal{D}_0$ generated by baseline foundation models.

Let $p(x\vert{}\theta)$ represent the parametric probability density function of text generation governed by a model parameterized by weights $\theta \in$ \[span_21](start_span)[span_21](end_span)[span_23](start_span)[span_23](end_span)Theta. The induced probability density function over the normalized stylometric space is denoted $p(\mathbf{z}\vert{}\theta)$. The information-theoretic geometry of this distribution is characterized by the Fisher Information Matrix (FIM) $\mathbf{F}(\theta) \in \mathbb{R}^{D \times D}$, whose components are defined as:

$$\mathbf{F}_{ij}(\theta) = \mathbb{E}_{x \sim p(x\vert{}\theta)}\left[\frac{\partial \log p(\mathbf{z}(x)\vert{}\theta)}{\partial z_i} \frac{\partial \log p(\mathbf{z}(x)\vert{}\theta)}{\partial z_j} \right] = \int_{\mathbb{R}^D} p(\mathbf{z}\vert{}\theta) \left(\frac{\partial \log p(\mathbf{z}\vert{}\theta)}{\partial z_i} \right) \left(\frac{\partial \log p(\mathbf{z}\vert{}\theta)}{\partial z_j} \right)$$

d\[span_26](start_span)[span_26](end_span)mathbf{z}

where $i, j \in \{1, 2, \dots, D\}$. The FIM induces a Riemannian metric tensor $g_{ij}(\mathbf{z}) = \mathbf{F}_{ij}(\mathbf{z})$ on the statistical manifold $\mathcal{M}$ of stylometric distributions. The infinitesimal distance ds^2 between two infinitely close stylometric parameterizations on $\mathcal{M}$ is given by the quadratic form:

$$ds^2 = g_{ij}(\mathbf{z}) dz^i dz^j = \sum_{i=1}^D \sum_{j=1}^D \mathbf{F}_{ij}(\mathbf{z}) dz_i dz_j$$

This Riemannian framework establishes that local distances on the stylometric manifold reflect

statistical discriminability: regions with high Fisher information correspond to stylometric directions where minor parameter shifts produce distinct, measurable changes in output style.
Analytical Construction of the Positive Semi-Definite Weighting Matrix
To distinguish genuine model lineage transformations from adversarial evasion attempts, the regularized Mahalanobis distance metric matrix $\mathbf{W} \in \mathbb{S}_+^D$ must maximize sensitivity to architectural variations while maintaining robustness against stylistic perturbations. The metric matrix is formulated as:

$$\mathbf{W} = \mathbf{\Sigma}_{\text{intra}}^{-1} + \mathbf{\Lambda}_{\text{invariance}}$$

where $\mathbf{\Sigma}_{\text{intra}} \in \mathbb{R}^{D \times D}$ represents the intra-family covariance matrix across non-adversarial fine-tuning, and $\mathbf{\Lambda}_{\text{invariance}} \in \mathbb{R}^{D \times D}$ is a regularizing invariant penalty matrix. Let $\mathcal{T}_{\text{family}}$ represent a set of standard model adaptations (such as domain-specific fine-tuning or quantization) applied to a target model root θ_0. The intra-family covariance is given by:

\mathbf{\Sigma}_{\text{intra}} = \mat[span_6](start_span)[span_6](end_span)hbb{E}_{\theta \sim \mathcal{T}_{\text{family}}}\left[(\mathbf{z}_{\theta} - \bar{\mathbf{z}})(\mathbf{z}_{\theta} - \bar{\mathbf{z}})^T \right]

where $\mathbf{z}_{\theta} = \mathbb{E}_{x \sim p(x\vert{}\theta)}[\mathbf{z}(x)]$ and $\bar{\mathbf{z}} = \mathbb{E}_{\theta}[\mathbf{z}_{\theta}]$.
To construct \mathbf{\Lambda}_{\te[span_12](start_span)[span_12](end_span)[span_18](start_span)[span_18](end_span)xt{invariance}}, consider an adversarial paraphrase transformation operator $\mathcal{A}_{\phi}: \mathcal{X} \to \mathcal{X}$ parameterized by attack vector $\phi \in \Phi$ designed to alter stylistic surface features without modifying core semantics. The expected adversarial feature perturbation covariance is defined as:

$$\mathbf{\Sigma}_{\text{adv}} = \mathbb{E}_{x \sim \mathcal{D}, \phi \sim \Phi}\left[(\mathbf{z}(\mathcal{A}_{\phi}(x)) - \mathbf{z}(x)) (\mathbf{z}(\mathcal{A}_{\phi}(x)) - \mathbf{z}(x))^T \right]$$

The diagonal entries of $\mathbf{\Lambda}_{\text{invariance}}$ are assigned inversely proportional to the variance induced by adversarial perturbations, penalized by feature-wise task-sensitivity gradients:

$$\mathbf{\Lambda}_{\text{invariance}} = \lambda \cdot \operatorname{diag}\left(\frac{\gamma_1}{\sigma_{\text{adv}, 1}^2 + \epsilon}, \frac{\gamma_2}{\sigma_{\text{adv}, 2}^2 + \epsilon}, \dots, \frac{\gamma_D}{\sigma_{\text{adv}, D}^2 + \epsilon} \right)$$

where $\sigma_{\text{adv}, i}^2 = [\mathbf{\Sigma}_{\text{adv}}]_{ii}$, $\gamma_i = \mathbb{E}_{\theta}\left[\left\vert{} \frac{\partial \mathcal{L}_{\text{task}}}{\partial z_i} \right\vert{} \right]$ measures the functional alignment of feature z_i with downstream task execution, $\lambda > 0$ is a global regularization scalar, and $\epsilon > 0$ prevents numerical instability.
This analytical construction guarantees that $\mathbf{W}$ is symmetric positive semi-definite ($\mathbf{W} \succeq 0$). Dimensions susceptible to low-cost adversarial manipulation (high $\sigma_{\text{adv}, i}^2$) receive attenuated weights, whereas dimensions linked to structural model execution (high [span_56](start_span)[span_56](end_span)[span_58](start_span)[span_58](end_span)\gamma_i and low intra-family drift) dominate the distance metric.

Manifold Deformation under Model Transformations

When a base model with parameters θ_0 undergoes parameter transformations

\mathcal{T}_\theta, the local metric tensor deforms from g_{ij}(\mathbf{z}) to g'_{ij}(\mathbf{z}). The nature of this deformation depends on the transformation class:

1. **Parameter-Efficient Fine-Tuning (LoRA)**: Low-Rank Adaptation updates weights via rank-constrained decomposition \Delta \theta = \mathbf{B}\mathbf{A}, where \mathbf{B} \in \mathbb{R}^{d_{\text{in}} \times r}, \mathbf{A} \in \mathbb{R}^{r \times d_{\text{out}}}, and r \ll \min(d_{\text{in}}, d_{\text{out}}). The deformed metric tensor satisfies:g'_{ij}(\mathbf{z}) = g_{ij}(\mathbf{z}) + \sum_{k=1}^r \frac{\partial g_{ij}(\mathbf{z})}{\partial \theta_k} (\mathbf{B}\mathbf{A})_[span_7](start_span)[span_7](end_span)k + \mathcal{O}(\Vert{}\mathbf{B}\mathbf{A}\Vert{}^2) Because \operatorname{rank}(\mathbf{B}\mathbf{A}) \le r, metric tensor deformation is strictly confined to an r-dimensional subspace of the tangent bundle T\mathcal{M}. Stylometric features aligned with the orthogonal complement \operatorname{ker}(\mathbf{B}\mathbf{A}) remain invariant, preserving ancestral lineage fingerprints.
2. **Full Parameter Fine-Tuning**: Full parameter optimization introduces unconstrained perturbations \mathbf{\Delta}_{\text{full}} \in \mathbb{R}^{\vert{}\theta\vert{}}. The transformation deforms the entire metric tensor field across all dimensions:g'_{ij}(\mathbf{z}) =[span_31](start_span)[span_31](end_span) g_{i[span_13](start_span)[span_13](end_span)[span_19](start_span)[span_19](end_span)j}(\mathbf{z}) + \nabla_\theta g_{ij}(\mathbf{z})^T \mathbf{\Delta}_{\text{full}} + \frac{1}{2} \mathbf{\Delta}_{\text{full}}^T \left(\nabla_\theta^2 g_{ij}(\mathbf{z}) \right) \mathbf{\Delta}_{\text{full}} This induces high-rank diffusion across the stylometric manifold, shifting both mean state vectors \boldsymbol{\mu} and local covariance profiles.
3. **Quantized Knowledge Distillation**: Quantization scales model weights to lower bit-widths alongside student model distillation \theta_S = \arg\min_\theta \mathcal{L}_{\text{KD}}(M_\theta, M_{\theta_0}). Quantization introduces bounded discretization noise \mathbf{N}_q, yielding metric tensor deformation:\mathbf{\Sigma}_{\text{distill}} = \mathbf{\Sigma}_0 + \mathbf{\Sigma}_q, \quad \mathbf{\Sigma}_q = \mathbb{E}[\mathbf{N}_q \mathbf{N}_q^T] The metric tensor transforms via an isotropic variance expansion: g'_{ij}(\mathbf{z}) = (\mathbf{I} + \mathbf{\Sigma}_0^{-1}\mathbf{\Sigma}_q)^{-1}_{ik} g_{kj}(\mathbf{z}), preserving macro-structural syntactic relationships while increasing feature variance.

Adversarial Robustness, Evasion Boundaries, and Feature Stability

Constrained Optimization Formulation of Evasion Boundaries

An adversary attempting to obfuscate model lineage seeks to generate a perturbed text x_{\text{adv}} = x + \Delta x that minimizes task execution loss \mathcal{L}_{\text{task}} while forcing the extracted stylometric vector \mathbf{z}(x_{\text{adv}}) within an \epsilon_{\text{budget}} distance threshold of a target false manifest \mathbf{z}_{\text{target}}, or pushing it beyond the detection boundary of the ancestral manifest \mathbf{z}_{\t[span_40](start_span)[span_40](end_span)[span_49](start_span)[span_49](end_span)ext{parent}}. This evasion boundary is formulated as a constrained optimization problem:

$$\min_{\Delta x \in \mathcal{X}_{\text{pert}}} \mathcal{L}_{\text{task}}(x + \Delta x) \quad \text{subject to} \quad d_{\mathbf{W}}(\mathbf{z}(x + \Delta x), \mathbf{z}_{\text{target}}) \le \epsilon_{\text{budget}}$$

where the regularized Mahalanobis distance $d_{\mathbf{W}}(\mathbf{z}_1, \mathbf{z}_2)$ is given by:

$$d_{\mathbf{W}}(\mathbf{z}_1, \mathbf{z}_2) = \sqrt{(\mathbf{z}_1 - \mathbf{z}_2)^T \mathbf{W} (\mathbf{z}_1 - \mathbf{z}_2)}$$

Using a first-order Taylor series approximation around the baseline output vector $\mathbf{z}(x)$, the perturbed feature vector is expressed via the feature Jacobian \mathbf{J}_{\[span_82](start_span)[span_82](end_span)mathbf{z}}(x) = \frac{\partial \mathbf{z}(x)}{\partial x} \in \mathbb{R}^{D \times \vert{}x\vert{}}:

$$\mathbf{z}(x + \Delta x) \approx \mathbf{z}(x) + \mathbf{J}_{\mathbf{z}}(x) \Delta x$$

Substituting this approximation into the distance constraint yields the local quadratic boundary condition:

$$\left(\mathbf{z}(x) + \mathbf{J}_{\mathbf{z}}(x) \Delta x - \mathbf{z}_{\text{target}} \right)^T \mathbf{W} \left(\mathbf{z}(x) + \mathbf{J}_{\mathbf{z}}(x) \Delta x - \mathbf{z}_{\text{target}} \right) \le \epsilon_{\text{budget}}^2$$

The adversarial perturbation energy ΔE_i required to shift a specific feature dimension z_i by a unit distance δ_i without degrading task efficiency is defined as:

\Delta E_i = \min_{\Delta x} \left\{ \Vert{}\Delta x\Vert{}_2^2 \;\middle\vert{}\; \left\vert{}[\mathbf{J}_{\mathbf{z}}(x) \Delta x]_i \right\vert{} = \delta_i, \, \nabla_x \mathcal{L}_{\te[span_84](start_span)[span_84](end_span)xt{task}}^T \Delta x \le \eta \right\}

where $\eta > 0$ represents the maximum acceptable task performance degradation budget.

Feature Volatility and Topological Stability Hierarchy

Stylometric feature dimensions in $\mathbf{\Psi}(X)$ exhibit non-uniform sensitivity to adversarial perturbations. Surface-level lexical statistics demonstrate low perturbation energy costs ($\Delta E_i \to 0$), making them vulnerable to automated synonym substitution and stylistic post-editing. Conversely, deep structural syntactic features and topological graph invariants require substantial perturbation energy ($\Delta E_i \gg 0$), rendering them stable against semantic-preserving evasion attacks.

Feature Category	Dimension Index Range	Feature Variable z_i	Mathematical Formulation	Perturbation Energy Cost (ΔE_i)	Adversarial Stability Index (S_i)
Lexical Richness	$z_1 - z_5$	Yule's Characteristic K	$K = 10^4 \cdot \frac{\left(\sum_{m=1}^{M} m^2 f_m \right) - N}{N^2}$	Very Low (0.12 - 0.25)	0.18 (Highly Volatile)
Surface Entropy	$z_6 - z_{10}$	Character/Punctuation Entropy H_p	$H_p = -\sum_{c \in \mathcal{C}_p} P(c) \log_2 P(c)$	Very Low (0.08 - 0.19)	0.12 (Highly Volatile)
Morphosyntactic	$z_{11} - z_{20}$	POS n-gram Frequencies	P(w_t \mid w_{t-1}, w_{t-2}) =	Moderate (0.45 - 0.62)	0.54 (Moderately Stable)

Feature Category	Dimension Index Range	Feature Variable z_i	Mathematical Formulation	Perturbation Energy Cost (ΔE_i)	Adversarial Stability Index (S_i)
			$\frac{C(w_{t-2}, w_{t-1}, w_t)}{C(w_{t-2}, w_{t-1})}$		
Syntactic Depth	z_{21} - z_{30}	Clause Subordination Ratio	$R_{\text{sub}} = \frac{N_{\text{subordinate}}}{N_{\text{main}} + N_{\text{subordinate}}}$	High (1.85 - 2.40)	0.82 (High Stability)
Dependency Topology	z_{31} - z_{40}	Graph Max Depth & Branching	$D_{\text{dep}} = \max_{v \in V_{\mathcal{G}}} \text{dist}(v_{\text{root}}, v)$	Very High (3.10 - 4.25)	0.91 (High Stability)
Persistent Homology	z_{41} - z_{50}	1D Betti Curve Integral	$I_{\beta_1} = \int_{0}^{\epsilon_{\max}} \beta_1(\epsilon) d\epsilon$	Extreme (5.80 - 7.50)	0.97 (Topologically Invariant)

The Adversarial Stability Index $S_i \in [0, 1]$ is computed as:
S_i = \frac{\Delta E_i}{\Delta E_[span_43](start_span)[span_43](end_span)[span_52](start_span)[span_52](end_span)i + \operatorname{Var}_{\phi \sim \Phi}\left([\mathbf{z}(\mathcal{A}_\phi(x))]_i \right)}
Surface metrics such as Yule's Characteristic K (measuring vocabulary richness where f_m is the count of words appearing m times in a text of N words) and punctuation entropy degrade under simple paraphrasing rules. In contrast, higher-order structural metrics—such as clause subordination ratios derived from context-free grammar (CFG) parse trees and dependency depth statistics—depend on the model's underlying cognitive planning mechanics. Altering these structures requires deep sentence restructures that alter the output's core semantics, raising task loss $\mathcal{L}_{\text{task}}$ beyond acceptable parameters.

Cybernetic Feedback and Drift Accumulation in Multi-Generation Model Forks

Discrete Markov State-Space Model for Recursive Derivatives

Consider a sequence of recursive derivative model forks $M_0 \to M_1 \to \dots \to M_k$, where each child model M_{k+1} is trained, instruction-tuned, or distilled using synthetic datasets $\mathcal{D}_{k+1} \sim p(x\vert{}\theta_k)$ generated by its immediate predecessor M_k. This multi-generational lineage is modeled as a discrete-time Markov state-space process defined over the normalized stylometric state vector $\mathbf{z}_k = \mathbf{z}(M_k) \in$

\mathbb{[span_68](start_span)[span_68](end_span)R}^D:
\mathbf{z}_{k+1} = \mathbf{A}_k \mathbf{z}_k + (\mathbf{I} - \mathbf{A}_k) \boldsymbol{\mu}_{\text{data}} + \boldsymbol{\eta}_k
where \mathbf{A}_k \in \mathbb{R}^{D \times D} represents the continuous state retention transition matrix at generation k, \boldsymbol{\mu}_{\text{data}} \in \mathbb{R}^D is the asymptotic attractor vector corresponding to the empirical center of gravity of the background training data distribution, \mathbf{I} \in \mathbb{R}^{D \times D} is the identity matrix, and \boldsymbol{\eta}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_k) represents an independent zero-mean Gaussian process noise vector with covariance matrix \mathbf{Q}_k \in \mathbb{R}^{D \times D} capturing stochastic optimization variance during fine-tuning.
By recursive expansion, the state vector at generation k is expressed as a function of the ancestral seed state \mathbf{[span_70](start_span)[span_70](end_span)z}_0:
\mathbf{z}_k = \left(\prod_{m=0}^{k-1} \mathbf{A}_m \right) \mathbf{z}_0 + \sum_{j=0}^{k-1} \left(\prod_{m=j+1}^{k-1} \mathbf{A}_m \right) (\mathbf{I} - \mathbf{A}_j) \boldsymbol{\mu}_{\text{data}} + \sum_{j=0}^{k-1} \left(\prod_{m=j+1}^{k-1} \mathbf{A}_m \right) \boldsymbol{\eta}_j
Assuming stationary transition dynamics (\mathbf{A}_k = \mathbf{A} and \mathbf{Q}_k = \mathbf{Q} for all k), this simplifies to:

\mathbf{z}_k = \mathbf{A}^k \mathbf{z}_0 + (\mathbf{I} - \mathbf{A}^k) \boldsymbol{\mu}_{\text{data}} + \sum_{j=0}^{k-1} \mathbf{A}^{k-1-j} \boldsymbol{\eta}_j

Asymptotic Stability and Stylometric Collapse Bounds

To evaluate lineage stability across extended derivative trees, evaluate the asymptotic expectation of the regularized Mahalanobis distance between generation k and ancestral root M_0:
\mathbb{E}\left[d_{\mathbf{W}}^2(\mathbf{z}_k, \mathbf{z}_0) \right] = \mathbb{E}\left[(\mathbf{z}_k - \mathbf{z}_0)^T \mathbf{W} (\mathbf{z}_k - \mathbf{z}_0) \right] = \operatorname{Tr}\left(\mathbf{W} \operatorname{Cov}(\mathbf{z}_k - \mathbf{z}_0) \right) + \Vert{}\mathbb{E}[\mathbf{z}_k] - \mathbf{z}_0\Vert{}_{\mathbf{W}}^2
where \Vert{}\math[span_85](start_span)[span_85](end_span)bf{v}\Vert{}_{\mathbf{W}}^2 = \mathbf{v}^T \mathbf{W} \mathbf{v}. The expectation of the state vector at generation k is:
\mathbb{E}[\mathbf{z}_k] = \mathbf{A}^k \mathbf{z}_0 + (\mathbf{I} - \mathbf{A}^k) \boldsymbol{\mu}_{\text{data}}
Let \rho(\mathbf{A}) = \max_i \{\vert{}\lambda_i(\mathbf{A})\vert{}\} denote the spectral radius of the state transition matrix \mathbf{A}.

1. **Stationary State Convergence (\rho(\mathbf{A}) < 1)**: If all eigenvalues of \mathbf{A} lie strictly inside the complex unit circle, then \lim_{k \to \infty} \mathbf{A}^k[span_74](start_span)[span_74](end_span) = \mathbf{0}. The expected state converges to the background data distribution center:\lim_{k \to \infty} \mathbb{E}[\mathbf{z}_k] = \boldsymbol{\mu}_{\text{data}} The asymptotic state covariance matrix \mathbf{P}_\infty = \lim_{k \to \infty} \operatorname{Cov}(\mathbf{z}_k) satisfies the discrete Lyapunov equation:\mathbf{P}_\infty = \mathbf{A} \mathbf{P}_\infty \mathbf{A}^T + \mathbf{Q} The asymptotic mean square distance bound evaluates to:\lim_{k \to \infty} \mathbb{E}\left[d_{\mathbf{W}}^2(\mathbf{z}_k, \mathbf{z}_0) \right] = \Vert{}\boldsymbol{\mu}_{\text{data}} - \mathbf{z}_0\Vert{}_{\mathbf{W}}^2 + \operatorname{Tr}(\mathbf{W} \mathbf{P}_\infty)
2. **Stylometric Collapse Dynamics**: When models recursively train on synthetic outputs

without variance-preserving constraints, the process noise covariance shrinks ($\mathbf{Q}_k \to \mathbf{0}$) along structural syntactic dimensions while the transition matrix eigenvalues collapse ($\rho(\mathbf{A}) \to 0$). This condition triggers *stylometric collapse*, wherein the system loses its unique structural markers ($\mathbf{P}_\infty \to \mathbf{0}$) and degenerates into an uninformative, low-entropy state. Under complete stylometric collapse, ancestral attribution breaks down as $d_{\mathbf{W}}(\mathbf{z}_k, \mathbf{z}_0) \to \Vert{}\boldsymbol{\mu}_{\text{data}} - \mathbf{z}_0\Vert{}_{\mathbf{W}}$, rendering derivative models indistinguishable from generic task background noise.

Homeostatic Control Mechanisms for OSSL Governance

To prevent lineage identity loss across multi-generational open-source forks, the OSSL governance protocol enforces a cybernetic homeostatic feedback mechanism. A dynamic transformation drift budget $\beta_{\text{budget}}(k)$ scales decay exponentially across lineage depth k:

\beta_{\text{budge[span_110](start_span)[span_110](end_span)t}}(k) = \beta_0 \cdot \gamma^k \quad \text{where } \gamma \in (0, 1)

where $\beta_0 > 0$ is the baseline single-generation drift allowance and γ is the attenuation coefficient.

For a candidate fork M_k to achieve valid lineage certification under OSSL v1.1, it must satisfy two homeostatic criteria:

1. **Generation-Bound Drift Constraint**: $d_{\mathbf{W}}(\mathbf{z}_k, \mathbf{z}_{k-1}) \le \beta_{\text{budget}}(k)$
2. **Cumulative Ancestral Horizon Constraint**: d_{\mathbf{W}}(\mathbf{z}_k, \mathbf{z}_0) \le \sum_{m=1}^k \beta_{\text{budget}}(m) = \beta_0 \frac{\gamma(1 - \gamma^k)}{1 - \gamma} < \frac{\beta_0 \gamma[span_112](start_span)[span_112](end_span)}{1 - \gamma}

If a model fork violates these bounds, the OSSL verification engine rejects deterministic lineage attribution, flagging the derivative as an unverified or severed branch.

Specification Enhancements for OSSL v1.1

Topological Data Analysis on Syntactic Dependency Graphs

OSSL v1.1 integrates Topological Data Analysis (TDA) on syntactic parse trees to extract invariant structural representations that resist surface text perturbations. Given an output text x, an automated dependency parser constructs a directed acyclic graph $\mathcal{G} = (V, E)$, where $V = \{v_1, v_2, \dots, v_n\}$ represents extracted tokens and $E \subseteq V \times V$ represents typed syntactic dependency relations.

To construct a persistent homology filtration, a symmetric metric space (V, d_D) is defined over the node set using the diffusion distance $d_D(v_i, v_j)$:

$$d_D(v_i, v_j) = W_1\left(P_t(v_i, \cdot), P_t(v_j, \cdot) \right)$$

where W_1 is the 1-Wasserstein metric and $P_t(v_i, \cdot)$ denotes the t-step random walk transition probability distribution starting from node v_i over dependency graph $\mathcal{G}$.

A Vietoris-Rips simplicial complex filtration $\mathcal{V}_\epsilon(\mathcal{G})$ is constructed over scale parameters $\epsilon \in [0, \epsilon_{\max}]$:

\mathcal{V}_\epsilon(\mathcal{G}) = \left\{ \sigma \subseteq V \;\middle\vert{}\; d_D(u, v) \le

$\epsilon, \, \forall u, v \in \sigma$
[span_127](start_span)[span_127](end_span)[span_129](start_span)[span_129](end_span)\right\}

As scale ϵ increases, structural transitions induce linear maps between persistence homology groups $H_p(\mathcal{V}_\epsilon; \mathbb{R})$ for dimensions $p \in \{0, 1\}$. This filtration yields a set of persistence pairs $\mathcal{D}_p = \{(b_m, d_m)\}_{m=1}^{K_p}$, where b_m and d_m mark the birth and death scales of topological connected components ($p=0$) and 1D structural loops ($p=1$).

From these persistence diagrams, OSSL v1.1 extracts four topological scalar invariants:

1. **Mean Persistence Lifetime (L_p)**: $L_p = \frac{1}{K_p} \sum_{m=1}^{K_p} (d_m - b_m)$
2. **Persistence Entropy ($H_{\text{pda}, p}$)**: $H_{\text{pda}, p} = -\sum_{m=1}^{K_p} p_m \log_2 p_m \quad \text{where } p_m = \frac{d_m - b_m}{\sum_{k=1}^{K_p} (d_k - b_k)}$
3. **Integrated Betti Curve (I_{β_1})**: [span_135](start_span)[span_135](end_span)[span_136](start_span)[span_136](end_span)$I_{\beta_1} = \int_0^{\epsilon_{\max}} \beta_1(\epsilon) d\epsilon$ where $\beta_1(\epsilon) = \dim H_1(\mathcal{V}_\epsilon; \mathbb{R})$ counts 1D topological cycles at scale ϵ.
4. **Persistent Hodge Laplacian Minimal Non-Zero Eigenvalue ($\lambda_{\min}(\Delta_1)$)**: Extracting the spectrum of the 1-dimensional Hodge Laplacian boundary operator $\Delta_1 = K_1^* K_1 + K_2 K_2^*$ yields higher-order connectivity metrics, where K_p represents the boundary matrix of dimension p.

Decentralized Multi-Parent Merkle Directed Acyclic Graphs

When derivative models merge knowledge from multiple parent models (e.g., combining model parameters or performing ensemble distillation across parents $\mathcal{P} = \{M_{p_1}, M_{p_2}, \dots, M_{p_N}\}$), the child stylometric state vector $\mathbf{z}_{\text{child}}$ reflects an interpolation of its parents' feature representations. Let $w_i \ge 0$ denote the fusion weight associated with parent M_{p_i}, such that $\sum_{i=1}^N w_i = 1$. The theoretical child centroid vector is modeled as:

$$\mathbf{z}_{\text{expected}} = \sum_{i=1}^N w_i \mathbf{z}_{p_i}$$

The merge residual innovation vector $\boldsymbol{\delta}_{\text{merge}} \in \mathbb{R}^D$ isolates unique fine-tuning shifts introduced during the model merge step:

$$\boldsymbol{\delta}_{\text{merge}} = \mathbf{z}_{\text{child}} - \mathbf{z}_{\text{expected}}$$

OSSL v1.1 encapsulates multi-parent merge events using a decentralized Merkle Directed Acyclic Graph (DAG) architecture. Each node in the DAG represents a discrete model state whose cryptographic envelope hash H_{node} is calculated as:

$$H_{\text{node}} = \text{SHA-256}\left(H(M_{p_1}) \parallel H(M_{p_2}) \parallel \dots \parallel H(M_{p_N}) \parallel w_1 \parallel w_2 \parallel \dots \parallel w_N \parallel \mathbf{z}_{\text{child}} \parallel \boldsymbol{\delta}_{\text{merge}} \right)$$

where $\parallel$ denotes byte-level concatenation, and $H(M_{p_i})$ represents the canonical manifest hash of parent i. This structure provides cryptographic provenance guarantees and structural traceability across complex model merge histories.

Algorithmic Implementation of the Adaptive Weighting Matrix Estimator

The Python pipeline implementation below constructs the positive semi-definite metric weighting matrix \mathbf{W} = \mathbf{\Sigma}_{\text{intra}}^{-1} + \mathbf{\Lambda}_{\text{invariance}} via pseudo-inverse covariance estimation and spectral projection.

```
import numpy as np
from scipy.linalg import pinv

def estimate_adaptive_weight_matrix(
    z_intra_samples: np.ndarray,
    z_adv_samples: np.ndarray,
    task_gradients: np.ndarray,
    lambda_reg: float = 1.0,
    epsilon: float = 1e-6
) -> np.ndarray:
    """
    Computes the regularized positive semi-definite weighting matrix
W.

    Parameters:
        z_intra_samples: Array of shape (N_intra, D) containing
z-score vectors
                         across intra-family model transformations.
        z_adv_samples:   Array of shape (N_adv, D, 2) containing
paired z-score
                         vectors (original, perturbed) under
adversarial paraphrasing.
        task_gradients:  Array of shape (D,) containing expected task
loss gradient
                         magnitudes per feature dimension |d_Ltask /
d_zi|.
        lambda_reg:      Global scalar tuning invariant penalty
weight.
        epsilon:         Numerical stability factor.

    Returns:
        W: Symmetric positive semi-definite matrix of shape (D, D).
    """
    N_intra, D = z_intra_samples.shape

    # Compute Intra-Family Covariance Matrix
    mu_intra = np.mean(z_intra_samples, axis=0)
    z_centered = z_intra_samples - mu_intra
    Sigma_intra = (z_centered.T @ z_centered) / (N_intra - 1)

    # Compute Regularized Inverse via Pseudo-Inverse
    Sigma_intra_inv = pinv(Sigma_intra, rcond=1e-15)

    # Compute Adversarial Perturbation Variances per Feature Dimension
    adv_diffs = z_adv_samples[:, :, 1] - z_adv_samples[:, :, 0]
```

```
    sigma_adv_sq = np.var(adv_diffs, axis=0)

    # Construct Invariant Penalty Matrix Lambda_invariance (Diagonal)
    gamma = np.abs(task_gradients)
    penalty_diag = gamma / (sigma_adv_sq + epsilon)
    Lambda_invariance = lambda_reg * np.diag(penalty_diag)

    # Synthesize Total Weighting Matrix W
    W = Sigma_intra_inv + Lambda_invariance

    # Enforce Exact Numerical Symmetry
    W = 0.5 * (W + W.T)

    # Spectral Projection to Guarantee Positive Semi-Definiteness (W
>= 0)
    eigenvalues, eigenvectors = np.linalg.eigh(W)
    eigenvalues_clipped = np.maximum(eigenvalues, 0.0)
    W_psd = eigenvectors @ np.diag(eigenvalues_clipped) @
eigenvectors.T

    return W_psd
```

Formal Schema Specification for OSSL Seed Version 1.1

The complete JSON schema specification updates the OSSL seed manifest standard to version 1.1, formally incorporating TDA persistent homology features, multi-parent Merkle DAG node attributes, and homeostatic validation parameters.


```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "OSSLStylometricSeedManifestV1_1",
  "type": "object",
  "required": [
    "manifest_version",
    "model_identifier",
    "cryptographic_envelope",
    "stylometric_state",
    "topological_invariants",
    "lineage_graph",
    "verification_parameters"
  ],
  "properties": {
    "manifest_version": {
      "type": "string",
      "enum": ["1.1"]
    },
    "model_identifier": {
      "type": "object",
```

```
      "required": ["model_name", "architecture_family",
"parameter_count_billions"],
      "properties": {
        "model_name": { "type": "string" },
        "architecture_family": { "type": "string" },
        "parameter_count_billions": { "type": "number", "minimum": 0.0
}
      }
    },
    "cryptographic_envelope": {
      "type": "object",
      "required": ["weights_sha256", "manifest_hash",
"signature_ecdsa_secp256k1"],
      "properties": {
        "weights_sha256": { "type": "string", "pattern":
"^[a-fA-F0-9]{64}$" },
        "manifest_hash": { "type": "string", "pattern":
"^[a-fA-F0-9]{64}$" },
        "signature_ecdsa_secp256k1": { "type": "string" }
      }
    },
    "stylometric_state": {
      "type": "object",
      "required": ["feature_dimension", "z_score_mean_vector",
"weighting_matrix_W"],
      "properties": {
        "feature_dimension": { "type": "integer", "const": 50 },
        "z_score_mean_vector": {
          "type": "array",
          "items": { "type": "number" },
          "minItems": 50,
          "maxItems": 50
        },
        "weighting_matrix_W": {
          "type": "array",
          "items": {
            "type": "array",
            "items": { "type": "number" },
            "minItems": 50,
            "maxItems": 50
          },
          "minItems": 50,
          "maxItems": 50
        }
      }
    },
    "topological_invariants": {
      "type": "object",
```

```
      "required": [
        "mean_persistence_lifetime_L0",
        "mean_persistence_lifetime_L1",
        "persistence_entropy_H0",
        "persistence_entropy_H1",
        "integrated_betti_1",
        "hodge_laplacian_min_eigenvalue"
      ],
      "properties": {
        "mean_persistence_lifetime_L0": { "type": "number" },
        "mean_persistence_lifetime_L1": { "type": "number" },
        "persistence_entropy_H0": { "type": "number" },
        "persistence_entropy_H1": { "type": "number" },
        "integrated_betti_1": { "type": "number" },
        "hodge_laplacian_min_eigenvalue": { "type": "number" }
      }
    },
    "lineage_graph": {
      "type": "object",
      "required": ["node_type", "parent_nodes",
"merge_fusion_weights", "residual_innovation_vector"],
      "properties": {
        "node_type": {
          "type": "string",
          "enum": ["root_seed", "single_parent_fork",
"multi_parent_merge"]
        },
        "parent_nodes": {
          "type": "array",
          "items": {
            "type": "object",
            "required": ["parent_manifest_hash",
"parent_weights_sha256"],
            "properties": {
              "parent_manifest_hash": { "type": "string", "pattern":
"^[a-fA-F0-9]{64}$" },
              "parent_weights_sha256": { "type": "string", "pattern":
"^[a-fA-F0-9]{64}$" }
            }
          }
        },
        "merge_fusion_weights": {
          "type": "array",
          "items": { "type": "number", "minimum": 0.0, "maximum": 1.0
}
        },
        "residual_innovation_vector": {
          "type": "array",
```

```
          "items": { "type": "number" },
          "minItems": 50,
          "maxItems": 50
        }
      }
    },
    "verification_parameters": {
      "type": "object",
      "required": ["generation_depth_k", "base_drift_budget_beta0",
"attenuation_gamma", "max_mahalanobis_threshold"],
      "properties": {
        "generation_depth_k": { "type": "integer", "minimum": 0 },
        "base_drift_budget_beta0": { "type": "number",
"exclusiveMinimum": 0.0 },
        "attenuation_gamma": { "type": "number", "exclusiveMinimum":
0.0, "maximum": 1.0 },
        "max_mahalanobis_threshold": { "type": "number",
"exclusiveMinimum": 0.0 }
      }
    }
  }
}
```

Conclusions

This analysis establishes a mathematical framework for tracking model lineage across open-source software distributions using intrinsic stylometric invariants. By constructing a regularized metric tensor \mathbf{W} rooted in Fisher Information manifold geometry and penalizing volatile surface dimensions, the system remains sensitive to structural parameter transformations while resisting low-cost adversarial evasion. Integrating Topological Data Analysis (TDA) on syntactic dependency graphs further stabilizes tracking against semantic-preserving paraphrasing.

The state-space drift models show that recursive derivative model forks undergo stylometric decay analogous to model collapse unless governed by dynamic homeostatic feedback loops. Incorporating multi-parent Merkle DAG structures and formal JSON schema specifications provides a scalable, decentralized approach to validating model provenance across the open-source model development ecosystem.

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